

AI-DRIVEN CLIMATE-AWARE DECISION SUPPORT AND RISK FORECASTING SYSTEMS FOR SRI LANKA

PROJECT ID : R26-IT-046



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Comments	Our Improvements
Reduce the scope to Colombo district.	Reduced the system scope and most processing components to only focus on Colombo District to make the project easier to manage and improve prediction accuracy.
Does not have much understanding of model training.	Tested multiple machine learning model training methods to identify the most accurate model for use in the flood prediction application.

The Problem and It's Impact

Problem

- People **do not receive** early flood warnings.
- People are **not aware** of dengue risk levels in their surrounding areas.
- Farmers **do not receive** specific advice related to weather and flood conditions.
- People find it difficult to identify **safe travel routes** during floods.



Impact

- Improves public safety
- Increases awareness of dengue-prone areas
- Supports farmers with location-based advice
- Helps people identify safer travel routes

Design Excellence

- Easy-to-use interface for flood information access
- Clear interactive flood risk map
- Accurate predictions using trained ML models
- Real-time alerts with updated data



IT22084422 | T.D.C.D INDUWARA

AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING

AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING

How the Solution Addresses the Sub-Problem

- Localized flood prediction system for Colombo District.
- Real-time flood risk mapping with clear visualization.
- Early warning alerts using live environmental data.
- Machine learning–based prediction for improved accuracy and response speed.

User Requirements Addressed by the Solution

- Early flood warning alerts for at-risk areas.
- Flood risk mapping to identify affected zones.
- Real-time updates on rainfall and river conditions.

AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING

Design Excellence / Contribution

- Developed and compared multiple machine learning models for flood prediction accuracy.
- Built a localized prediction system focused on Colombo District data.
- Integrated real-time rainfall, river level, and elevation data into the model pipeline.
- Designed an end-to-end system from data processing to map-based visualization and alerts.

User Feedback on Prototype

- Users found the system easy to understand and navigate.
- Helpful for quick flood-related information and decision-making.
- Improved awareness of flood risks and safety actions.
- Useful for getting timely updates and guidance during floods .



IT22100702 | LAKSHAN H.M.M

AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM

AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM

How the Solution Addresses the Sub-Problem

- Predicts future dengue risk levels for each Colombo District division.
- Uses machine learning for accurate dengue outbreak forecasting.
- Integrates dengue, weather, and mosquito breeding data.
- Provides real-time and future 4-week dengue risk prediction.

User Requirements Addressed by the Solution

- Early dengue warning alerts for high-risk areas.
- Division-wise dengue risk monitoring and mapping.
- Real-time updates using weather API data.
- Easy-to-understand Normal, Warning, and Critical risk levels.

AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM

Design Excellence / Contribution

- Combines multiple datasets for higher prediction accuracy.
- Uses advanced ML models: XGBoost, Random Forest, and LSTM.
- Includes dengue breeding cycle logic in prediction features.
- Interactive color-based dengue risk visualization system.

User Feedback on Prototype

- Users found the risk map simple and easy to understand.
- Early warning predictions were considered highly useful.
- Division-wise risk predictions improved local awareness.
- Users preferred real-time updates and mobile accessibility.



IT22299666 | ATHUKORALA C.J

CLIMATE-INFORMED PADDY ADVISORY SYSTEM

CLIMATE-INFORMED PADDY ADVISORY SYSTEM

How the Solution Addresses the Sub-Problem

- Predicts expected paddy yield using machine learning
- Detects weather-related risks such as flood and drought.
- Recommends suitable fertilizer amounts and timings
- Provides AI-generated farming advice in Sinhala and English.

Prototype User Requirements Addressed by the Solution

- Farmers can input district, season, and crop details.
- System generates yield predictions and fertilizer recommendations.
- Provides weather forecasts and risk alerts
- Supports a simple mobile-friendly bilingual interface.

CLIMATE-INFORMED PADDY ADVISORY SYSTEM

Design Excellence / Contribution

- Integrates machine learning with agricultural advisory logic.
- Uses advanced ML models: XGBoost, Random Forest.
- Combines weather, soil, and historical yield datasets.
- Provides real-time smart farming decision support.

User Feedback on Prototype

- Users liked the simple and user-friendly interface.
- Sinhala language support was highly appreciated.
- Farmers found risk alerts and fertilizer advice useful.
- Users suggested adding disease detection and more districts.



IT22553232 | ABHISHEKA K.K.R.H

AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM

AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM

How the Solution Addresses the Sub-Problem

- AI-based safe route prediction for Colombo District.
- Detects floods, traffic congestion and blocked roads.
- Provides real-time road risk analysis and alerts.
- Generates safer alternative routes using live data and AI predictions.

User Requirements Addressed by the Solution

- Safe route suggestions to avoid risky roads.
- Real-time alerts for floods, rain, traffics.
- Live road condition monitoring and updates
- Easy navigation with alternative route recommendations.

AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM

Design Excellence / Contribution

- Combines AI prediction with smart navigation.
- Uses real-time weather and traffic integration.
- Predicts climate-based road block risks using machine learning.
- Dynamic route recalculation for safer travel.

User Feedback on Prototype

- Users found safe route suggestions useful.
- Alternative routes helped reduce delays.
- Map visualization was easy to understand.

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Commercialization / Sustainability

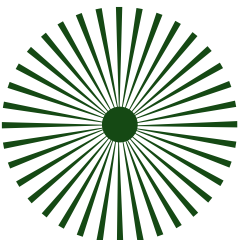
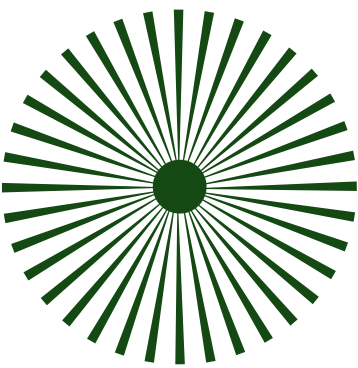
- Government disaster management system use and scalable across Sri Lanka and other countries.
- Serves disaster authorities, local governments, farmers, and public in high-risk areas.
- Targets disaster response, agriculture, and public safety sectors for long-term sustainability.

Cost Recovery

- Government funding and institutional deployment contracts.
- Subscription-based advanced features and analytics services.
- Partnerships with agriculture and insurance sectors for risk assessment and data services.

What is Unique

- All-in-one flood prediction and mapping system.
- Real-time ML-based updates with location-specific insights.
- Combines flood forecasting, risk mapping, and decision support in one platform.



THANK YOU ...

